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SUMMARY

M.S. ECE engineer across the full AI-systems stack: optimizes CUDA/Triton kernels and profiles LLM inference at the bottom. Designs agent infrastructure, retrieval, and evaluation on top. Shipped kernel-fusion and GPU-profiling work for LLM workloads, an MCP memory layer for AI coding assistants, and autonomous multi-agent systems that write code, run experiments, and govern themselves.

SKILLS

- **GPU & Performance:** CUDA, Triton (kernel DSL), PyTorch, kernel fusion, mixed precision (fp16), NVIDIA Nsight Systems/Compute, GPU performance profiling, roofline/memory-bandwidth analysis, quantization, gem5, McPAT
- **AI Systems:** LLM inference profiling, FlashAttention/SDPA, KV-cache decode, multi-agent orchestration, MCP server design, Claude Code / Agent SDK, agent evaluation, governance/anti-forgery
- **Retrieval & Memory:** Hybrid search, reciprocal-rank fusion, cross-encoder reranking, sqlite-vec, SQLite FTS5, knowledge graphs
- **Languages & Infra:** Python, C++, CUDA, Docker, CI/CD, Slurm

EDUCATION

University of Wisconsin–Madison

M.S. Electrical and Computer Engineering

Madison, USA

Sep 2024 – Dec 2025

- Relevant Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

Vellore Institute of Technology

B.Tech in Electronics and Communication Engineering

Amaravati, India

2020 – 2024

PROJECTS

Triton vs cuBLAS LLM Kernel Benchmarking

Triton, CUDA, PyTorch, NVIDIA Nsight

Dec 2025 – Present

GitHub

- Fused linear+bias+GeLU into one **Triton** kernel for **up to 1.73×** lower p50 latency than **cuBLAS** on small-batch FFN projections, removing an intermediate **HBM** round-trip and a kernel launch.
- Benchmarked **76 LLM-shaped** GEMM kernels across Triton and cuBLAS on an **A100** (p50/p90 latency, throughput, bandwidth, tail jitter) to map where fusion wins. The tiled Triton GEMM sustains **215 TFLOP/s**, **69% of fp16 tensor-core peak**.

Edge LLM GPU Profiling on Jetson Orin Nano

CUDA, NVIDIA Nsight Systems, PyTorch, FlashAttention

Feb 2026

GitHub

- Profiled **attention** and **KV-cache decode** on a Jetson Orin Nano (edge GPU for local LLM-agent deployment), sustaining **~15.4K tokens/sec** single-stream FlashAttention decode and **~7.5 TFLOP/s** peak fp16 GEMM.
- Measured fused **SDPA/FlashAttention** at **~5.6×** over naive attention with Nsight Systems and traced a per-step decode kernel-launch-overhead pattern.

bert: Long-Context Memory for AI Coding Assistants (MCP)

May 2026 – Present

Python, MCP, sqllite-vec, sentence-transformers

GitHub

- Built **bert**, a local plugin (**MCP server**) that gives AI coding assistants (Claude Code, Cursor, Codex) a searchable memory of an entire codebase via **hybrid retrieval** (dense vector + keyword search, reranked), pulling only the relevant files into the model's context instead of overflowing its context window.
- Root-caused a silent bug in the retrieval fusion that had pinned answer accuracy near-random (**0.10**) and restored it to **0.85**. On the BEIR scifact IR benchmark, the dense+BM25 hybrid scores **0.684 nDCG@10** (published BM25: 0.665).
- Showed retrieval is the only workable path once a project exceeds the context window: on a **3M-token** corpus (3× a **1M** window), full-context is infeasible and truncation scores **0.00**, while bert answers at **0.75** accuracy on a flat **~3.3K** tokens per query.

Autonomous ML Research Lab (31-Agent System)

Mar 2026 – Present

Python, C17, Apple Metal, Claude API (Opus 4.8)

GitHub

- Built a **31-agent** autonomous system that runs an end-to-end **ML research loop** on a single 18 GB laptop: the agents plan experiments, write and run the code, and review each other's results, with a human only setting direction.
- Their core deliverable is a **from-scratch deep-learning training engine** in **C17** and **Apple Metal** (hand-written GPU kernels, a Mixture-of-Experts transformer, 4-bit quantization-aware training) with **no PyTorch or TensorFlow**.
- Designed the integrity layer that keeps autonomous agents honest: pre-registration, phase gates, and **cryptographic sign-off verification**, added after an agent forged 5 approvals, with **no recurrence over 23 runs**.

HIVE: Autonomous Multi-Agent Software Org

Mar 2026 – Present

Rust, libp2p, Python, Claude Code

- Built and run **HIVE**, an autonomous software org of **45 role-based agents** (Director, Evaluator, Red Team) that has completed **~203 build cycles** on a single product. Authored the governance layer that coordinates them: signed operator values, a tiered reversibility-based decision protocol, and JSON hand-off contracts between agents.
- Directed the agents to build a **66K-line Rust** system, a privacy-first **peer-to-peer reasoning engine**: a forward-chaining rules/inference engine (alpha-network **Rete** + truth-maintenance), a **16-module differential-privacy** layer that protects user data (Rényi-DP, k-anonymity), and a **libp2p** networking stack (Kademlia DHT, Gossipsub), gated by **1,300+ tests**.

ML-Guided CUDA Kernel Configuration

2025

Python, PyTorch, CUDA, Slurm

GitHub

- Trained a PyTorch MLP runtime surrogate (12 GEMM/launch features) that predicts CUDA kernel execution time at **R2 = 0.96** (MAE 0.018 ms), replacing exhaustive autotuning with argmin-over-grid config selection.
- Validated predicted block/grid configs to **within ~5% of the exhaustively-measured optimum** (median across workloads), with benchmarking and CUDA event-timing automated on Slurm.

MI300X Memory-System Characterization & gem5 Modeling

2025

HIP, gem5, McPAT, Python

- Characterized the **AMD MI300X** memory hierarchy with HIP microbenchmarks (vL1D ~130, L2 ~230, HBM ~630–760-cycle latencies, plus **4.6–4.8 TB/s** write bandwidth) and calibrated full-system **gem5** to **within 5% MAPE** of measured vL1D latency on the dominant access stride, down from 89%.
- Compared in-order and out-of-order CPU models in gem5 across SAXPY/DAXPY/IAXPY for CPI, latency, and power, with **McPAT** stats-to-power extraction.